

29.5.2020

Exercise 2

1: Study how to build a PLC program using TIA Portal. This exercise is just using Boolean information, so you are using only Boolean addresses like M0.0, I0.1 and Q1.0. M stands for Memory, I for Input and Q for output.

2: Build a very simple program: There are three inputs (I0.0 Open, I0.1 Close and I0.2 Lock). Inputs I0.0 and I0.1 are monostable pulses (they are true only when someone presses that button). Lock signal is bistable (it keeps in one position until someone changes a position). There is one output (Q0.0 Door). The door actuator is built so that when the actuator gets a true signal from PLC, the door is open. When it doesn't get that signal, door closes. Note there must be a memory (SR -gate) in program because the door must be open longer than only time when someone presses open button.

3: Remember also make a function call into main program (OB1).

4: Check by simulator that your program works. If it doesn't work, try to fix it and simulate it again. When it works, make a short video about your program in action (simulate different cases shortly).

5: Return your video as trained in exercise 1 into right returning box in Moodle.

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